MIL-S-36963A
23 September 1976
SUPERSEDING
MIL-S-36963 (DSA-DM)
24 August 1973

MILITARY SPECIFICATION

STIRRER-HOT PLATE, MAGNETIC, LABORATORY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of magnetic stirrer-hot plate for use in laboratory procedures, capable of stirring and heating simultaneously, or of doing either operation alone.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproof, Waterproof, Flexible.
MIL-M-7298	- Manual, Commercial, Technical Equipment.
MIL-L-10547	- Liners, Case and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.
MIL-B-13292	- Barrier Material, Waterproofed, Flexible, all Temperatures, Heat-Sealed.
MIL-B-22191	- Barrier Materials, Transparent, Flexible, Heat-Sealable.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C73-11	Dimensions of Plugs and Receptacles -- General Purpose, 125 Volts, 15 Amperes, 2-Pole, 3-Wire, Grounding Type.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018.)

THE CHEMICAL RUBBER COMPANY

Handbook of Chemistry and Physics

(Application for copies should be addressed to The Chemical Rubber Company, 10901 Cranwood Parkway, Cleveland, Ohio 44128.)

UNDERWRITERS' LABORATORIES, INC.

UL 499	Electric Heating Appliances.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, L.I., NY 11746; 207 East Ohio Street, Chicago, IL 60611; and 1655 Scott Boulevard, Santa Clara, CA 95050.)

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U.S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Fire and casualty hazards.

3.1.1 The contractor shall submit proof, prior to delivery, that the items to be delivered under the contract conform to the applicable requirements of UL 499. Proof shall consist of one of the following:

a. UL label or UL listing of the product, provided the contractor agrees to have Underwriters' Laboratories, Inc. forward a copy of the UL report directly to the contracting officer, upon the latter's request.

b. Laboratory report form a qualified independent laboratory. If an independent laboratory is used, it must be adequately equipped and competent to determine compliance with UL standards. The qualification of such laboratory shall be determined by the Government which reserves the right to require a facilities report and to conduct a facilities survey. Any laboratory report submitted shall include actual test methods, data taken, and results of tests. Detailed findings of examinations necessary to determine compliance shall also be included. The test report shall also have a section that describes the test sample in such a way that the report can be used to insure identity of production items with the test sample. This section shall be supplemented with construction information such as materials used, spacings between electrical components, types of bushings, photographs, wiring diagrams, sketches and parts list, all appropriately identified.

3.1.2 Compliance with above paragraph does not excuse compliance with all other requirements of the contract and specifications.

3.2 Materials. All materials used shall be suitable for the purposes intended and shall be as hereinafter indicated.

3.3 Components.

* 3.3.1 Top plate. The top plate shall be fabricated from Corning Glass Works' PYROCERAM Brand Crystalline Material 9608, or equal. The top plate shall be not less than 1/8-inch thick, and shall have at least a 5 x 7 inch rectangular top surface with a 7/8-inch minimum skirted overhang around the periphery of the top surface. The top plate shall be electrically non-conductive and shall have a glazed finish which shall be smooth, hard, scratch-resistant, and chemically inert. The manufacturer's normal commercial color finish shall be acceptable. All exposed corners and edges of the top plate shall be rounded.

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3.3.2 Base. The base shall be fabricated from 1/16-inch minimum aluminum. The bottom of the base shall be essentially rectangular with an area of 38-45 square inches with no side less than 5-1/2 inches. The base shall have suitable cutouts for the line cord and for mounting the electric control components. All exposed corners and edges of the base shall be rounded. The outside surfaces of the base shall have a suitable epoxy finish, which shall be smooth, hard, and scratch-resistant. The manufacturer's normal commercial color finish shall be acceptable. The bottom of the base shall have a removable corrosion-resistant plate to permit access to the internal components. The plate shall have a suitable ventilation opening. The base shall be supported by rubber or plastic feet, which shall be located at each corner.

3.3.3 Motor assembly. The motor assembly shall not require lubrication and shall consist of a shaded pole, high torque variable speed motor, having a magnet and fan connected to the drive shaft. The motor shall be suitable for operation on 115 volts, 50-60 Hz, AC supply lines. When operating on 50 Hz, AC, a 5/6 reduction in motor speed from that at 60 Hz, AC operation shall be acceptable.

3.3.4 Speed control. The speed control shall be a wire-wound rheostat with on-off positions, and shall be used to control the speed of the motor in the range of 100-950 revolutions per minute (rpm.). The rheostat shall be supplied with a graduated reference dial that is also permanently marked to indicate the "OFF" position and positions for increased speed of the motor. The dial shall have a suitable index for the dial position, or the index may be indicated on the base to which the dial and rheostat are mounted. The speed control shall be supplied with a pilot light circuit and suitable identification marking to indicate that the rheostat is in the "ON" position. The changes in motor speed shall be stepless from minimum to maximum speed.

3.3.5 Heating element. The heating element shall be a nickel-chromium ribbon wire that is secured on a suitable supporting plate which is both heat-resistant and electrically non-conducting.

3.3.6 Heater control. The heater control shall be an adjustable switch which controls the input power to the heating element by effecting the intermittent operation of the input current to the heating element (see 3.3.5). The heater control shall have an adjustable cam and bimetal thermostat type element to control the intermittent duty cycle of the heater current. The degree of heating for the hot plate shall be determined by the timing cycle of the bimetal element. The changes in temperature control shall be stepless from minimum to maximum. The heater control shall be supplied with a reference dial that is adjustable in clockwise and counter-clockwise directions. Permanent markings around the dial shall indicate the "OFF" position and the relative positions for increasing or decreasing the temperature of the hot plate. The dial or supporting base shall have a suitable index for positioning of the dial. The heater control shall be supplied with a pilot light circuit and suitable identification marking to indicate that the heater control is in the "ON" position.

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3.3.7 Line cord. A flexible line cord and plug, of one-piece construction, shall be provided with three 18 AWG conductors that terminate in a 3-blade male plug that is described in ANSI C73-11. The exposed length of line cord and plug shall be not less than 59 inches.

3.3.8 Stirring bar. A 1- by 5/16-inch thick or a 1 - by 3/8-inch thick teflon-coated magnetic stirring bar shall be supplied with the stirrer-hot plate apparatus.

3.4 Construction. The base shall house the motor and electrical controls, and shall support the top plate and heater assembly which shall be top-mounted. The heater assembly shall be rigidly positioned directly under the top plate and covered by an aluminum sheet with a circular opening to accommodate the magnetic bar on the motor drive shaft. The overhang of the top plate shall overlap the opening in the top of the base. Suitable ventilation openings shall be provided between the base and heater assembly. The motor assembly shall be positioned within the housing so that the drive shaft is perpendicular to the plane of the heater assembly, and the magnetic bar rotates close to the heater assembly in a plane parallel to the heater assembly. The speed and heater controls shall be securely attached to the front of the base. The line cord shall be connected to the electrical controls through a cutout in the back of the base. Strain relief shall be provided at the cutout to secure the line cord and prevent abrasion to the line cord insulation. The assembled unit shall have maximum overall dimensions of 7-3/4 long x 7 deep x 5-1/2 high (inches). When placed on a flat surface, the stirrer-hot plate shall be free of rocking and wobbling.

3.5 Performance. The heating element and motor operation shall be capable of independent and simultaneous operation and shall operate on 110-120 volt, 50/60 Hz, AC supply lines. With the motor and temperature controls at maximum, at 60 Hz operation, the stirrer-hot plate power consumption shall be approximately 600 watts. The pilot lights shall indicate when the controls are in the "ON" position. When tested in accordance with 4.4.1, each stirrer-hot plate shall operate without failure.

3.5.1 Stirring control. The stirrer-hot plate shall be capable of stirring liquids in the viscosity range of 1-1500 centipoise. The stirring action shall be accomplished by the stirring bar immersed in a vessel holding the liquid to be stirred. The vessel shall be composed of non-magnetic properties. When the vessel is placed on top of the hot plate over the motor assembly, the stirring action shall be accomplished by the motion of the stirring bar which is magnetically driven by the action of the rotating magnet on the variable speed motor. When tested in accordance with 4.4.2, the pilot light shall be on when the speed control is in the "ON" position; the motor action shall not cause the stirrer-hot plate to creep; the stirring bars shall attain speeds for the liquids (see 6.1) shown in Table I.

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TABLE I. RPM/Viscosity

Liquid	RPM
Distilled water at 20°C.	100 to 950
Glycerin (U.S.P.) at 30°C.	900 min.
Glycerin (U.S.P.) at 20°C.	500 min.

3.5.2 Heating control. When tested in accordance with 4.4.3, the pilot light shall indicate that the heater control is in the "ON" position, and the top plate shall attain a minimum temperature of 750° Fahrenheit (°F.) within 30 minutes. In addition, the hot plate shall show no evidence of permanent discoloration due to heating. A hot plate that is discolored while in a heated condition shall be acceptable provided it returns to its original color when cool.

3.5.3 Dielectric strength. There shall be no electrical break--down when the stirrer-hot plate is subjected to the dielectric test specified in 4.4.4.

3.6 Identification marking. Each stirrer-hot plate shall be permanently and legibly marked with the NSN, contract order number, and electrical characteristics. Marking secured by adhesives shall not be allowed.

3.7 Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract. In addition, the contractor shall furnish 15 copies of the approved technical manual to the contracting officer for distribution. Should the supplier have funished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the contracting officer. Waiver of submission of 15 additional copies shall be at the option of the Government.

3.8 Workmanship. Workmanship shall be first class throughout. The stirrer-hot plate shall be free from defects which detract from their appearance or may impair their serviceability. The stirrer-hot plate shall be free of sharp edges and dents. All internal electric wiring shall be adequately secure to prevent breaking off at terminations. No soldering shall result in cold solder joints. Any mechanical terminations shall be adequate to prevent breaking off or slipping off.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examination and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

* 4.2.1 For examination. Sampling for examination shall be conducted in accordance with Table II and MIL-STD-105.

TABLE II. For examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-1	2.5

4.2.2 For tests. Tests shall be conducted in accordance with Table III.

TABLE III. For tests

Characteristic	Requirement	Test procedure	Inspection level
Operation	3.5	4.4.1	100%
Stirring control	3.5.1	4.4.2	*
Heating control	3.5.2	4.4.3	*
Dielectric strength	3.5.3	4.4.4	100%

* One unit per lot. Acceptance number shall be zero.

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4.3 Examination. The stirrer-hot plate shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects shown in Table IV. Examination shall not be restricted to the classified possible defects listed in the table.

TABLE IV. Calssification of defects

Categories	Defects
Major	
101	Component parts of stirrer-hot plate not usable and not properly secured to unit.
102	Unit not free of cracks and distortion.
103	Unit not free of wobbling and rocking when placed on a flat surface.
104	Unit not free of sharp edges and burrs.
105	Hot plate surface not smooth or not free of permanent discoloration.
106	Line cord not adequately secured within housing.
107	Line cord not free of cuts, tears and abrasion.
108	Blades of plug (line plug) not properly aligned.
109	Component part(s) missing from assembled unit.
110	Internal wiring not adequately secured to terminations.
Minor	
201	Identification marking not complete, legible and permanent.
202	Finish on base not free of scratches, chipping and blisters.
203	Unit not free of dirt, grease and foreign matter.

4.3.2 Dimensional examination. The stirrer-hot plate shall be examined for defects in dimensions. Any dimension not within the limits specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Operation. A stirring magnet shall be placed in a vessel containing a suitable liquid for stirring. With the line plug connected and the vessel placed over the center of the top plate, the speed control shall be turned from minimum to maximum and the motor action and magnetic bar stirring action shall be checked for proper operation. The heating controls shall be adjusted for increasing temperature, and suitable means shall be used to determine if the hot plate temperature is rising.

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4.4.2 Speed control. Five hundred-milliliter (ml.) beakers shall be filled with 250 ml. of the following liquids:

a. Distilled water at 20° Centigrade (°C.)
b. Glycerin (U.S.P.) solutions at 30°C.
c. Glycerin (U.S.P.) solution at 20°C.

A stirring bar magnet (see 3.3.8) shall be completely immersed in the liquid in each beaker. With the line plug connected and a beaker placed over the center of the top plate, the speed control shall be turned from minimum to maximum. The speed of the stirring bar magnet at maximum setting (and minimum setting for distilled water) shall be determined by use of a stroboscopic light.

4.4.3 Temperature control. With the line plug connected, the temperature control shall be turned to maximum setting. The temperature at the center of the top plate shall be measured by using suitable thermocouples. After the surface of the hot plate is in a cool condition, it shall be examined for evidence of discoloration.

4.4.4 Dielectric strength. With the line plug disconnected and the control switches in the "OFF" positions, the stirrer-hot plate shall be subjected to a 60 Hz, AC potential gradually increasing to 1000 volts, that is applied between the current-carrying parts and the metal enclosure, and maintained at maximum value for a period of 1 minute.

4.5 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requriements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. Packaging. Packaging shall be Level A or C as specified (see 6.2).

5.1 Unit of issue. One stirrer-hot plate, as specified, constitutes one unit of issue.

5.1.1 Level A.

5.1.1.1 Unit package. Each unit shall be packaged in accordance with MIL-P-116, method IC-2. As an alternate, 4 mil polyethylene barrier material may be used in lieu of MIL-B-121, MIL-B-13239 or MIL-B-22191 barrier material as specified in 3.6.5.1.2 of MIL-P-116.

5.1.2 Level C. Units shall be packaged in accordance with contractor's commercial practice which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

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5.2.1 Level A.

5.2.1.1 Exterior container. Six unit packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification. Fiberboard boxes shall be waterproofed as specified in the appendix of PPP-B-636.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.2 Level B.

5.2.2.1 Exterior container. Six unit packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract.

5.3 Marking.

5.3.1 Interior packages. Each unit package shall be marked in accordance with MIL-STD-129. Contract number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

* 6.1 Testing liquids. The indicated liquids represent a range of viscosities up to 1500 centipoise. Based on the Handbook of Chemistry and Physics, the viscosity of each liquid is considered as follows:

Distilled water at 20°C.	1,002 cp.
Glycerin (U.S.P.) at 20°C.	1,490 cp.
Glycerin (U.S.P.) at 30°C.	629 cp.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.3 This specification does not include all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the type procured to meet military requirements.

6.4 This specification covers the following item listed in the Federal Supply Catalog:

National Stock Number	Item Identification
6640-00-113-8336	STIRRER-HOT PLATE, MAGNETIC, LABORATORY, 115 volt, 50/60 Hz, AC

6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6640-0987

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